

```
In [1]: %pip install numpy pandas matplotlib seaborn
```

```
Requirement already satisfied: numpy in C:\Users\Dell\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (2.5.0)
Requirement already satisfied: pandas in C:\Users\Dell\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (3.0.3)
Requirement already satisfied: matplotlib in C:\Users\Dell\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (3.11.0)
Requirement already satisfied: seaborn in C:\Users\Dell\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (0.13.2)
Requirement already satisfied: python-dateutil>=2.8.2 in C:\Users\Dell\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: tzdata in C:\Users\Dell\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from pandas) (2026.2)
Requirement already satisfied: contourpy>=1.0.1 in C:\Users\Dell\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in C:\Users\Dell\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in C:\Users\Dell\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from matplotlib) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in C:\Users\Dell\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from matplotlib) (1.5.0)
Requirement already satisfied: packaging>=20.0 in C:\Users\Dell\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from matplotlib) (26.2)
Requirement already satisfied: pillow>=9 in C:\Users\Dell\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from matplotlib) (12.2.0)
Requirement already satisfied: pyparsing>=3 in C:\Users\Dell\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from matplotlib) (3.3.2)
Requirement already satisfied: six>=1.5 in C:\Users\Dell\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

```
In [2]: # import python Libraries
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt # visualizing data
%matplotlib inline
import seaborn as sns
```

```
In [6]: # import csv file
```

```
df = pd.read_csv(r'C:\Users\Dell\Downloads\Python_Diwali_Sales_Analysis-main\Python_Diwali_Sales_Analysis-main\Diwali Sales Data.csv', encoding= 'unicode_escape')
```

```
In [7]: df.shape
```

```
Out[7]: (11251, 15)
```

```
In [8]: df.head(12)
```

```
Out[8]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th>
</th> <th>User_ID</th> <th>Cust_name</th> <th>Product_ID</th> <th>Gender</th> <th>Age Group</th> <th>Age</th> <th>Marital_Status</th> <th>State</th>
<th>Zone</th> <th>Occupation</th> <th>Product_Category</th> <th>Orders</th> <th>Amount</th> <th>Status</th> <th>unnamed1</th> </thead> <tbody> <th>0</th>
<th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> <th>9</th> <th>10</th> <th>11</th> </tbody> <tbody></tbody>
```

1002903	Sanskriti	P00125942	F	26-35	28	0	Maharashtra	Western	Healthcare	Auto	1	23952.00	NaN	NaN
1000732	Kartik	P00110942	F	26-35	35	1	Andhra Pradesh	Southern	Govt	Auto	3	23934.00	NaN	NaN
1001990	Bindu	P00118542	F	26-35	35	1	Uttar Pradesh	Central	Automobile	Auto	3	23924.00	NaN	NaN
1001425	Sudevi	P00237842	M	0-17	16	0	Karnataka	Southern	Construction	Auto	2	23912.00	NaN	NaN
1000588	Joni	P00057942	M	26-35	28	1	Gujarat	Western	Food Processing	Auto	2	23877.00	NaN	NaN
1000588	Joni	P00057942	M	26-35	28	1	Himachal Pradesh	Northern	Food Processing	Auto	1	23877.00	NaN	NaN
1001132	Balk	P00018042	F	18-25	25	1	Uttar Pradesh	Central	Lawyer	Auto	4	23841.00	NaN	NaN
1002092	Shivangi	P00273442	F	55+	61	0	Maharashtra	Western	IT Sector	Auto	1	NaN	NaN	NaN
1003224	Kushal	P00205642	M	26-35	35	0	Uttar Pradesh	Central	Govt	Auto	2	23809.00	NaN	NaN
1003650	Ginny	P00031142	F	26-35	26	1	Andhra Pradesh	Southern	Media	Auto	4	23799.99	NaN	NaN
1003829	Harshita	P00200842	M	26-35	34	0	Delhi	Central	Banking	Auto	1	23770.00	NaN	NaN
1000214	Kargatis	P00119142	F	18-25	20	0	Andhra Pradesh	Southern	Retail	Auto	2	23752.00	NaN	NaN

```
In [9]: df.info()
```

```

<class 'pandas.DataFrame'>
RangeIndex: 11251 entries, 0 to 11250
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   User_ID                11251 non-null  int64
1   Cust_name              11251 non-null  str
2   Product_ID            11251 non-null  str
3   Gender                 11251 non-null  str
4   Age Group              11251 non-null  str
5   Age                    11251 non-null  int64
6   Marital_Status         11251 non-null  int64
7   State                  11251 non-null  str
8   Zone                   11251 non-null  str
9   Occupation             11251 non-null  str
10  Product_Category       11251 non-null  str
11  Orders                 11251 non-null  int64
12  Amount                 11239 non-null  float64
13  Status                  0 non-null      float64
14  unnamed1               0 non-null      float64
dtypes: float64(3), int64(4), str(8)
memory usage: 1.3 MB

```

```

In [ ]: #drop unrelated/blank columns
df.drop(['Status', 'unnamed1'], axis=1, inplace=True)

```

```

In [15]: #check for null values
pd.isnull(df).sum()

```

```

Out[15]: User_ID                0
Cust_name                0
Product_ID              0
Gender                  0
Age Group               0
Age                    0
Marital_Status          0
State                   0
Zone                    0
Occupation              0
Product_Category        0
Orders                  0
Amount                 12
dtype: int64

```

```

In [16]: df.shape

```

```

Out[16]: (11251, 13)

```

```

In [17]: # drop null values
df.dropna(inplace=True)

```

```
In [18]: df.shape
```

```
Out[18]: (11239, 13)
```

```
In [19]: pd.isnull(df).sum()
```

```
Out[19]: User_ID      0
Cust_name      0
Product_ID     0
Gender         0
Age Group      0
Age           0
Marital_Status 0
State         0
Zone         0
Occupation     0
Product_Category 0
Orders        0
Amount        0
dtype: int64
```

```
In [22]: # change data type
df['Amount'] = df['Amount'].astype('int')
```

```
In [23]: df['Amount'].dtypes
```

```
Out[23]: dtype('int64')
```

```
In [24]: df.columns
```

```
Out[24]: Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',
               'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',
               'Orders', 'Amount'],
              dtype='str')
```

```
In [25]: #rename column
df.rename(columns={'Marital_Status': 'Marriage Status'})
```

```
Out[25]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th>
</th> <th>User_ID</th> <th>Cust_name</th> <th>Product_ID</th> <th>Gender</th> <th>Age Group</th> <th>Age</th> <th>Marriage Status</th> <th>State</th>
<th>Zone</th> <th>Occupation</th> <th>Product_Category</th> <th>Orders</th> <th>Amount</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th>
<th>4</th> <th>...</th> <th>11246</th> <th>11247</th> <th>11248</th> <th>11249</th> <th>11250</th> </tbody> <tbody></tbody>
```

1002903	Sanskriti	P00125942	F	26-35	28	0	Maharashtra	Western	Healthcare	Auto	1	23952
1000732	Kartik	P00110942	F	26-35	35	1	Andhra Pradesh	Southern	Govt	Auto	3	23934
1001990	Bindu	P00118542	F	26-35	35	1	Uttar Pradesh	Central	Automobile	Auto	3	23924
1001425	Sudevi	P00237842	M	0-17	16	0	Karnataka	Southern	Construction	Auto	2	23912
1000588	Joni	P00057942	M	26-35	28	1	Gujarat	Western	Food Processing	Auto	2	23877
...	...	...	...	...	...	...	...	...	...	...	...	...
1000695	Manning	P00296942	M	18-25	19	1	Maharashtra	Western	Chemical	Office	4	370
1004089	Reichenbach	P00171342	M	26-35	33	0	Haryana	Northern	Healthcare	Veterinary	3	367
1001209	Oshin	P00201342	F	36-45	40	0	Madhya Pradesh	Central	Textile	Office	4	213
1004023	Noonan	P00059442	M	36-45	37	0	Karnataka	Southern	Agriculture	Office	3	206
1002744	Brumley	P00281742	F	18-25	19	0	Maharashtra	Western	Healthcare	Office	3	188

<p>11239 rows × 13 columns</p>

```
In [26]: # describe() method returns description of the data in the DataFrame (i.e. count, mean, std, etc)
df.describe()
```

```
Out[26]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th>
</th> <th>User_ID</th> <th>Age</th> <th>Marital_Status</th> <th>Orders</th> <th>Amount</th> </thead> <tbody> <th>count</th> <th>mean</th> <th>std</th>
<th>min</th> <th>25%</th> <th>50%</th> <th>75%</th> <th>max</th> </tbody> <tbody></tbody>
```

1.123900e+04	11239.000000	11239.000000	11239.000000	11239.000000
1.003004e+06	35.410357	0.420055	2.489634	9453.610553
1.716039e+03	12.753866	0.493589	1.114967	5222.355168
1.000001e+06	12.000000	0.000000	1.000000	188.000000
1.001492e+06	27.000000	0.000000	2.000000	5443.000000
1.003064e+06	33.000000	0.000000	2.000000	8109.000000
1.004426e+06	43.000000	1.000000	3.000000	12675.000000
1.006040e+06	92.000000	1.000000	4.000000	23952.000000

```
In [28]: # use describe() for specific columns
df[['Age', 'Orders', 'Amount']].describe()
```

```
Out[28]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th>
</th> <th>Age</th> <th>Orders</th> <th>Amount</th> </thead> <tbody> <th>count</th> <th>mean</th> <th>std</th> <th>min</th> <th>25%</th> <th>50%</th>
<th>75%</th> <th>max</th> </tbody> <tbody></tbody>
```

11239.000000	11239.000000	11239.000000
35.410357	2.489634	9453.610553
12.753866	1.114967	5222.355168
12.000000	1.000000	188.000000
27.000000	2.000000	5443.000000
33.000000	2.000000	8109.000000
43.000000	3.000000	12675.000000
92.000000	4.000000	23952.000000

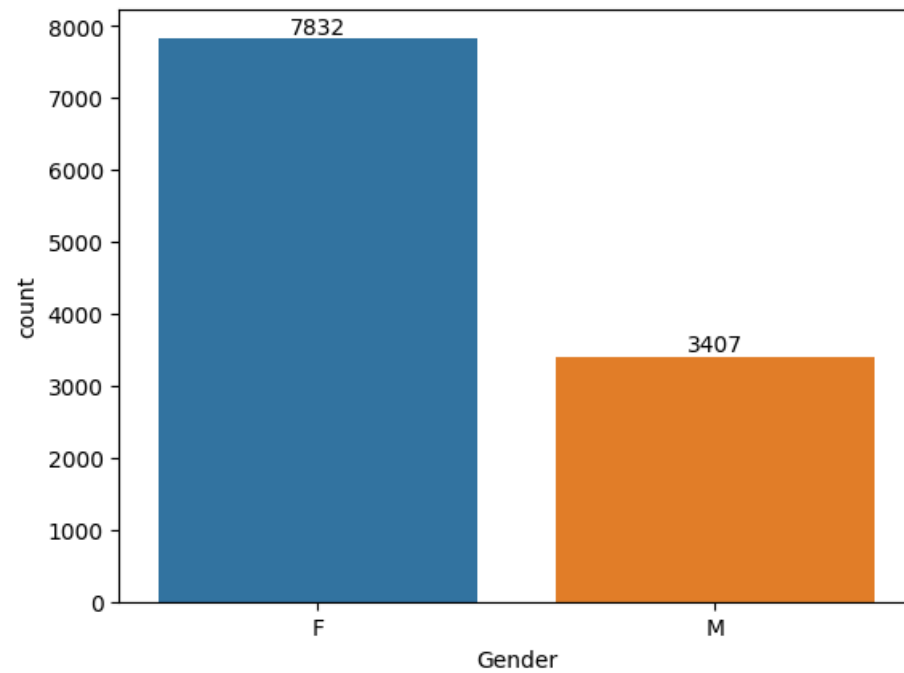
<h1 id="Exploratory-Data-Analysis">Exploratory Data Analysis</h1>

<h3 id="Gender">Gender</h3>

```
In [15]: # plotting a bar chart for Gender and it's count
```

```
ax = sns.countplot(x = 'Gender',data = df)

for bars in ax.containers:
    ax.bar_label(bars)
```

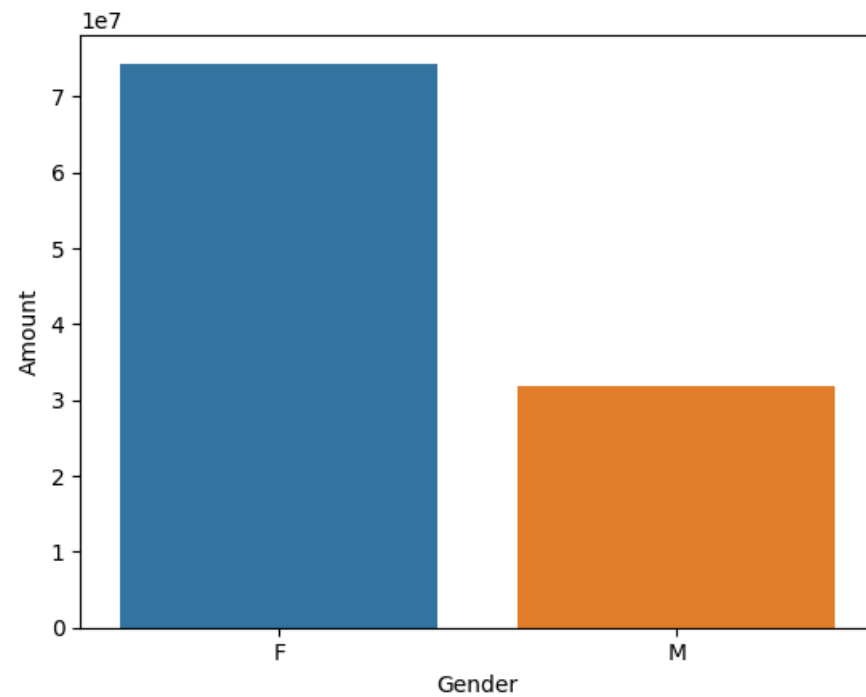


```
In [16]: # plotting a bar chart for gender vs total amount

sales_gen = df.groupby(['Gender'], as_index=False)['Amount'].sum().sort_values(by='Amount', ascending=False)

sns.barplot(x = 'Gender',y= 'Amount' ,data = sales_gen)
```

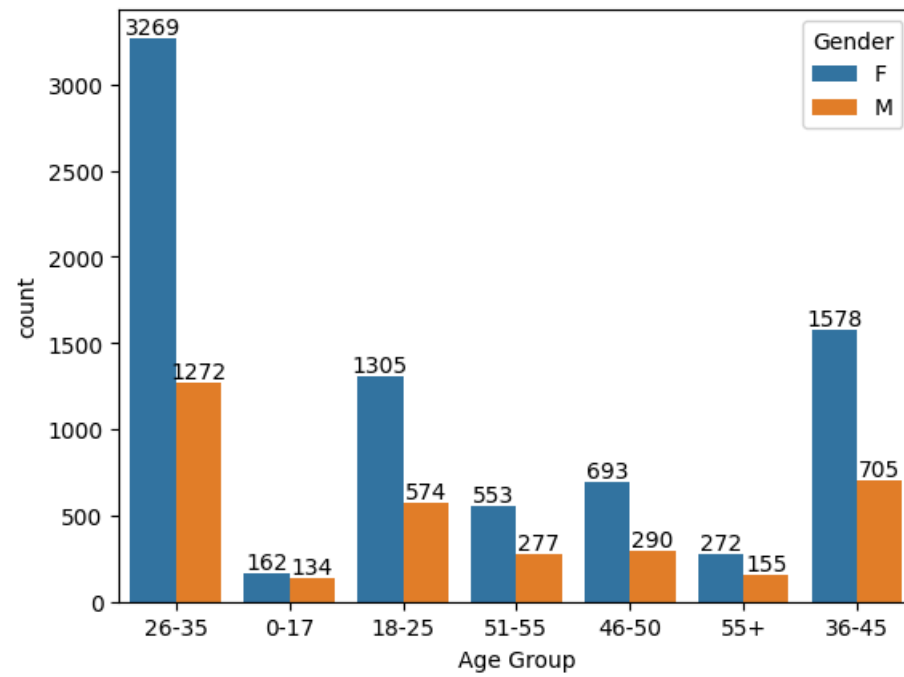
```
Out[16]: <Axes: xlabel='Gender', ylabel='Amount'>
```



<p>From above graphs we can see that most of the buyers are females and even the purchasing power of females are greater than men</p>

<h3 id="Age">Age</h3>

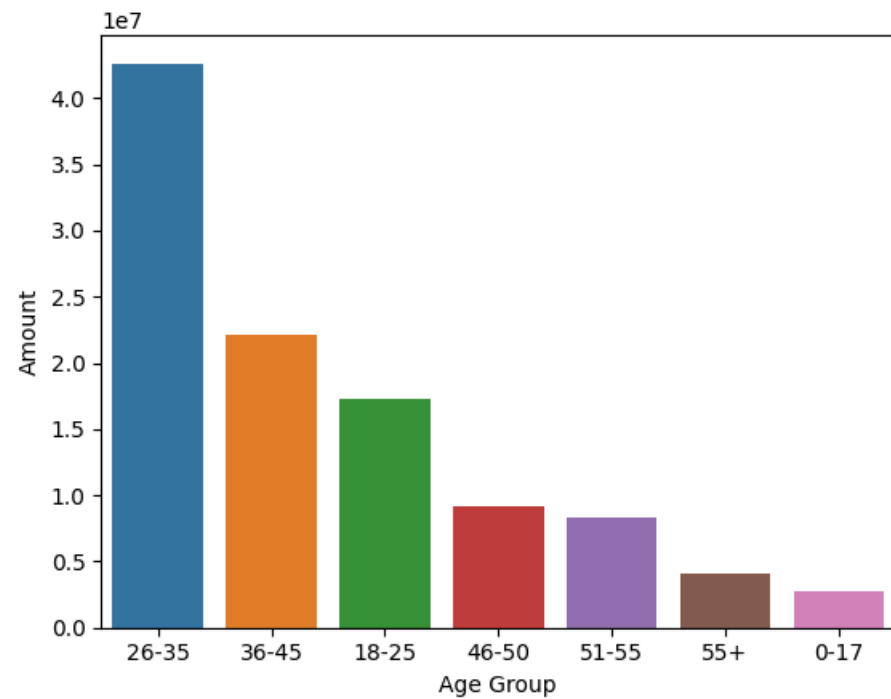
```
In [17]: ax = sns.countplot(data = df, x = 'Age Group', hue = 'Gender')
for bars in ax.containers:
    ax.bar_label(bars)
```

```
In [18]: # Total Amount vs Age Group
sales_age = df.groupby(['Age Group'], as_index=False)['Amount'].sum().sort_values(by='Amount', ascending=False)

sns.barplot(x = 'Age Group', y= 'Amount' ,data = sales_age)
```

```
Out[18]: <Axes: xlabel='Age Group', ylabel='Amount'>
```



<p>From above graphs we can see that most of the buyers are of age group between 26-35 yrs female</p>

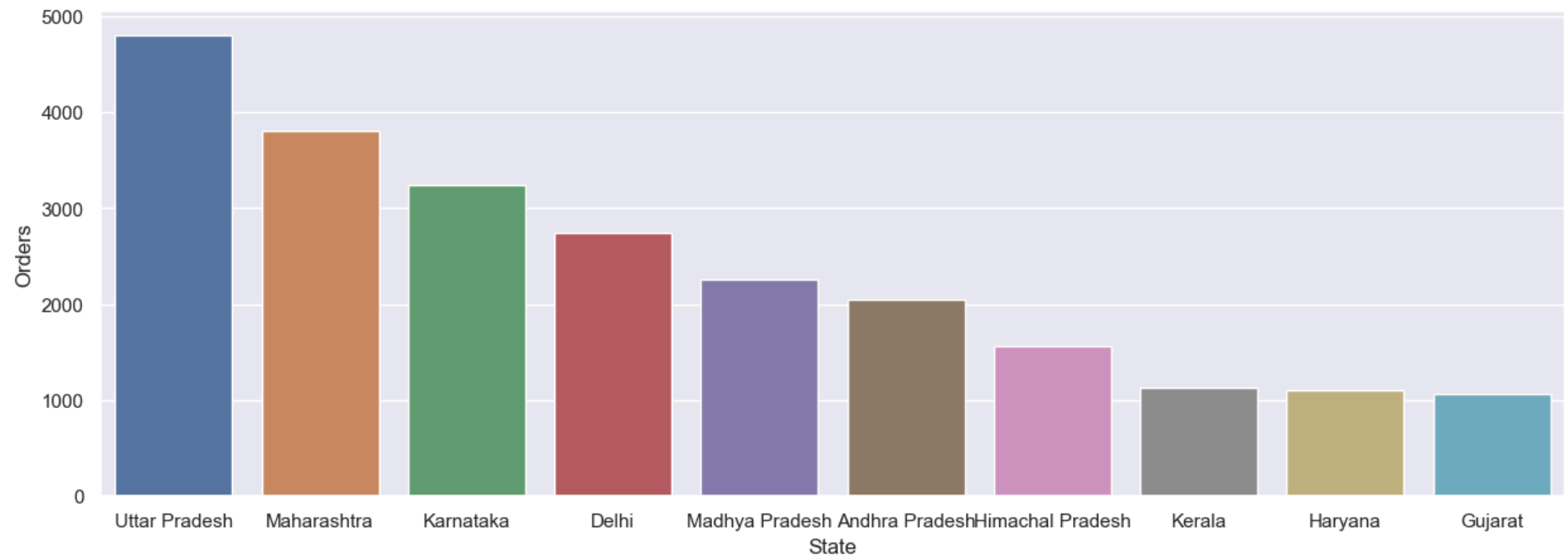
<h3 id="State">State</h3>

```
In [19]: # total number of orders from top 10 states

sales_state = df.groupby(['State'], as_index=False)['Orders'].sum().sort_values(by='Orders', ascending=False).head(10)

sns.set(rc={'figure.figsize':(15,5)})
sns.barplot(data = sales_state, x = 'State',y= 'Orders')
```

Out[19]: <Axes: xlabel='State', ylabel='Orders'>

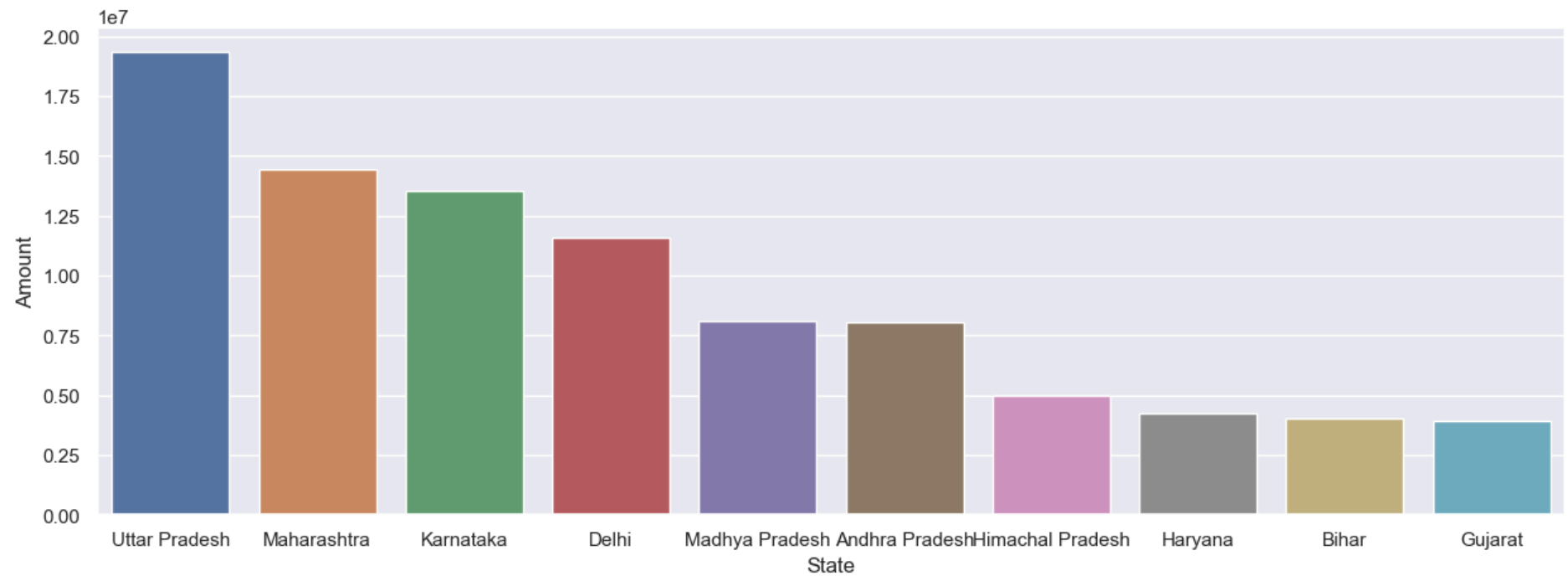


```
In [20]: # total amount/sales from top 10 states

sales_state = df.groupby(['State'], as_index=False)['Amount'].sum().sort_values(by='Amount', ascending=False).head(10)

sns.set(rc={'figure.figsize':(15,5)})
sns.barplot(data = sales_state, x = 'State',y= 'Amount')
```

```
Out[20]: <Axes: xlabel='State', ylabel='Amount'>
```

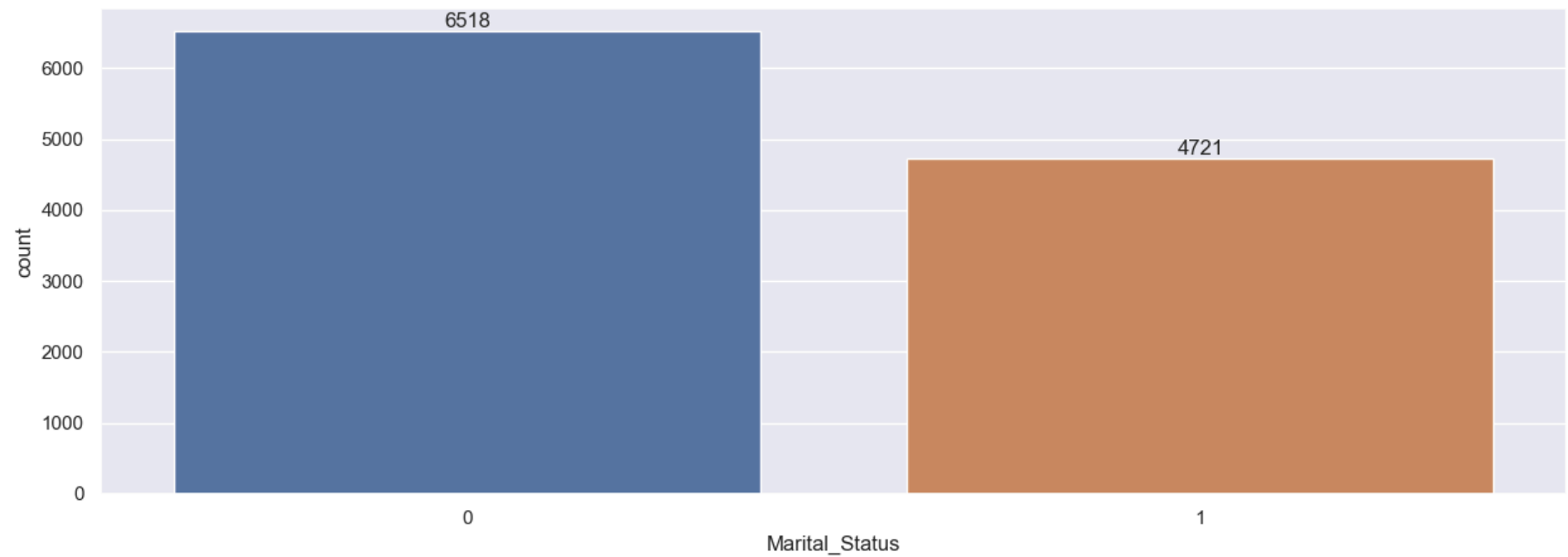


<p>From above graphs we can see that most of the orders & total sales/amount are from Uttar Pradesh, Maharashtra and Karnataka respectively</p>

<h3 id="Marital-Status">Marital Status</h3>

```
In [21]: ax = sns.countplot(data = df, x = 'Marital_Status')

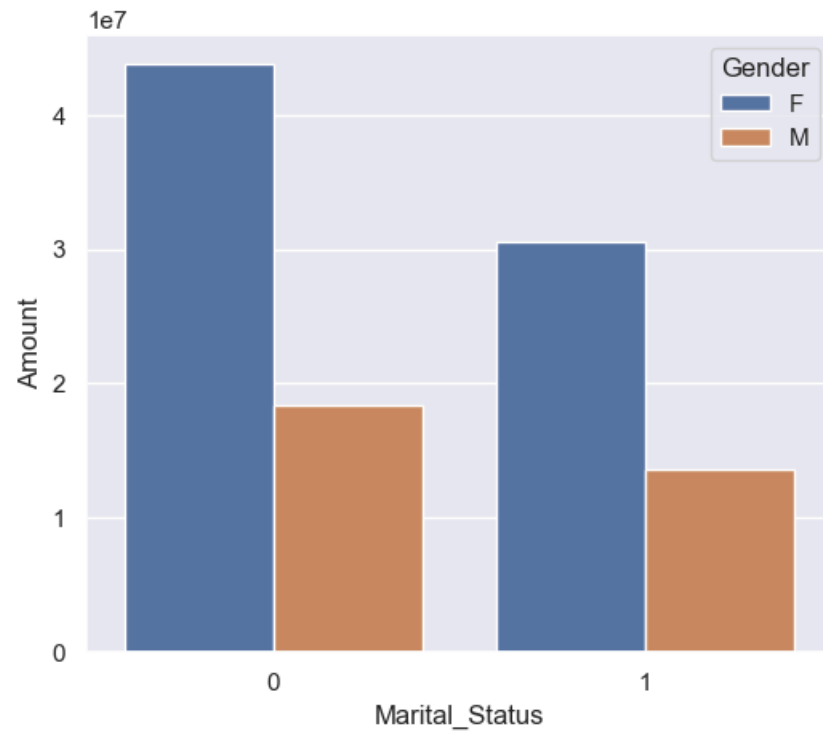
sns.set(rc={'figure.figsize':(7,5)})
for bars in ax.containers:
    ax.bar_label(bars)
```



```
In [22]: sales_state = df.groupby(['Marital_Status', 'Gender'], as_index=False)['Amount'].sum().sort_values(by='Amount', ascending=False)

sns.set(rc={'figure.figsize':(6,5)})
sns.barplot(data = sales_state, x = 'Marital_Status', y= 'Amount', hue='Gender')
```

```
Out[22]: <Axes: xlabel='Marital_Status', ylabel='Amount'>
```

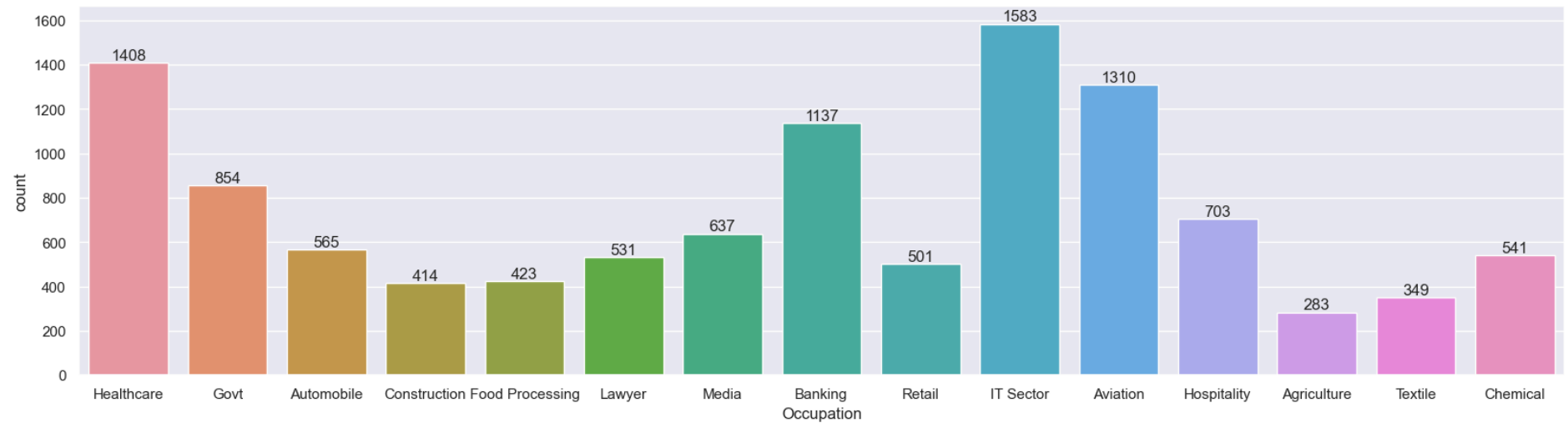


<p>From above graphs we can see that most of the buyers are married (women) and they have high purchasing power</p>

<h3 id="Occupation">Occupation</h3>

```
In [23]: sns.set(rc={'figure.figsize':(20,5)})
ax = sns.countplot(data = df, x = 'Occupation')

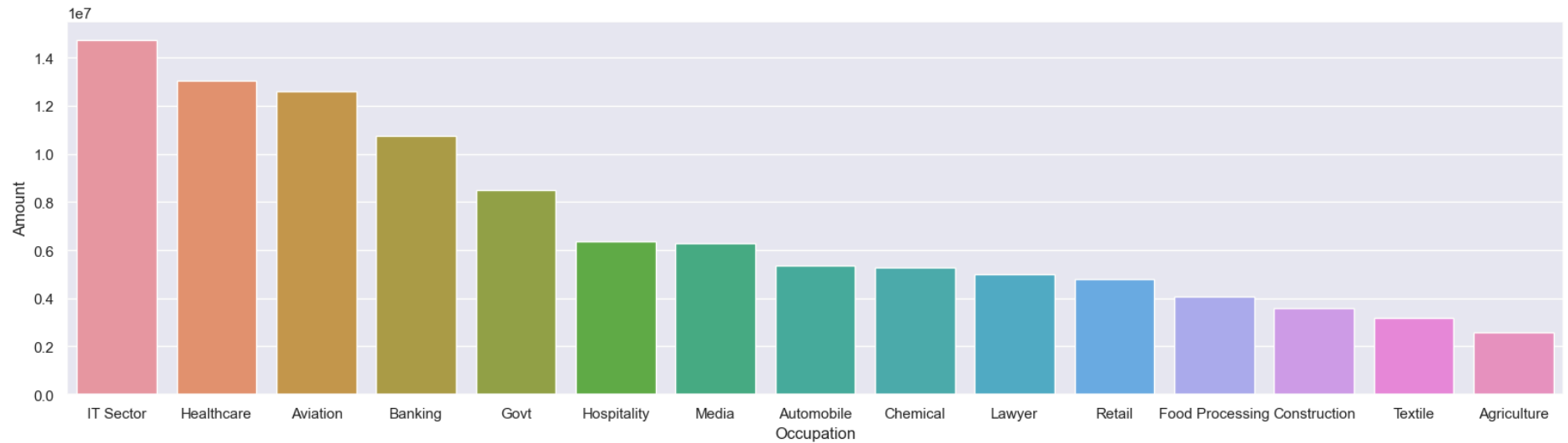
for bars in ax.containers:
    ax.bar_label(bars)
```



```
In [24]: sales_state = df.groupby(['Occupation'], as_index=False)['Amount'].sum().sort_values(by='Amount', ascending=False)

sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data = sales_state, x = 'Occupation',y= 'Amount')
```

Out[24]: <Axes: xlabel='Occupation', ylabel='Amount'>

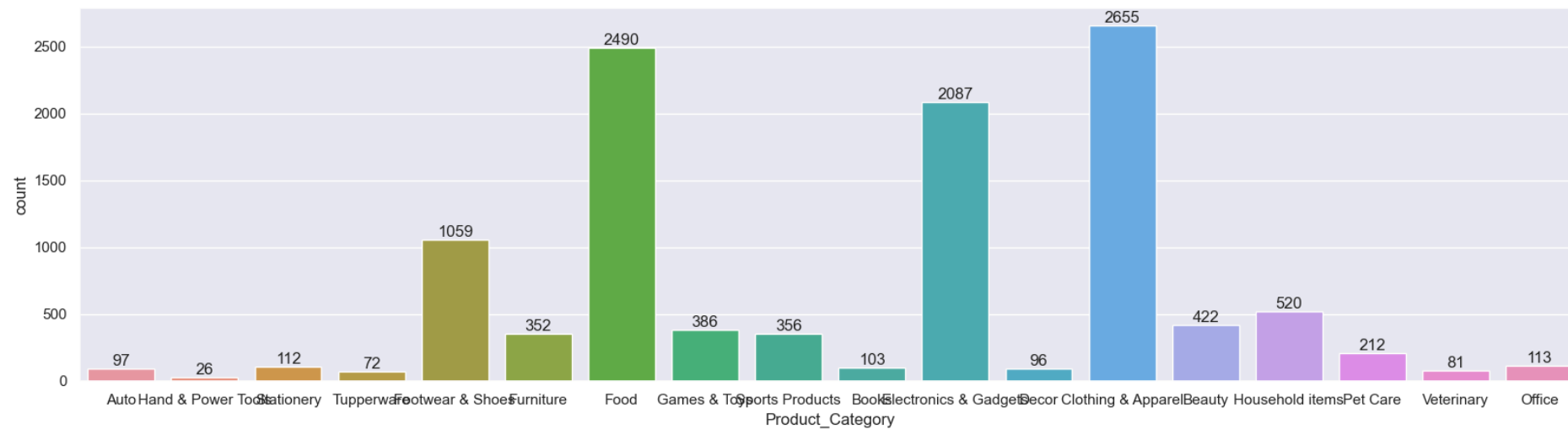


<p>From above graphs we can see that most of the buyers are working in IT, Healthcare and Aviation sector</p>

<h3 id="Product-Category">Product Category</h3>

```
In [25]: sns.set(rc={'figure.figsize':(20,5)})
ax = sns.countplot(data = df, x = 'Product_Category')

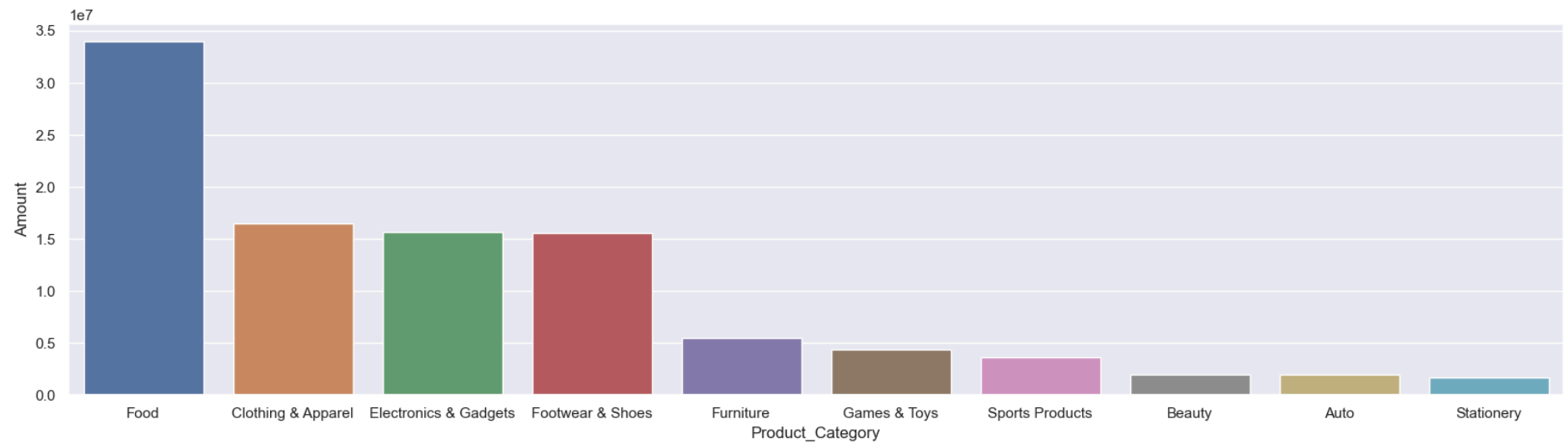
for bars in ax.containers:
    ax.bar_label(bars)
```



```
In [26]: sales_state = df.groupby(['Product_Category'], as_index=False)['Amount'].sum().sort_values(by='Amount', ascending=False).head(10)

sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data = sales_state, x = 'Product_Category', y= 'Amount')
```

```
Out[26]: <Axes: xlabel='Product_Category', ylabel='Amount'>
```

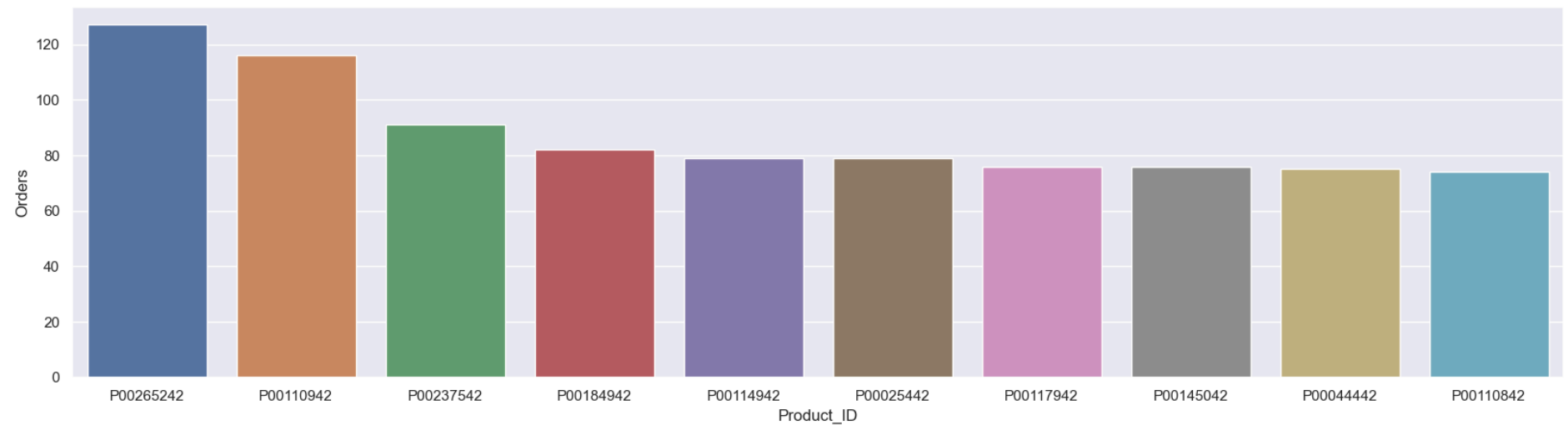



<p>From above graphs we can see that most of the sold products are from Food, Clothing and Electronics category</p>

```
In [27]: sales_state = df.groupby(['Product_ID'], as_index=False)['Orders'].sum().sort_values(by='Orders', ascending=False).head(10)

sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data = sales_state, x = 'Product_ID',y= 'Orders')
```

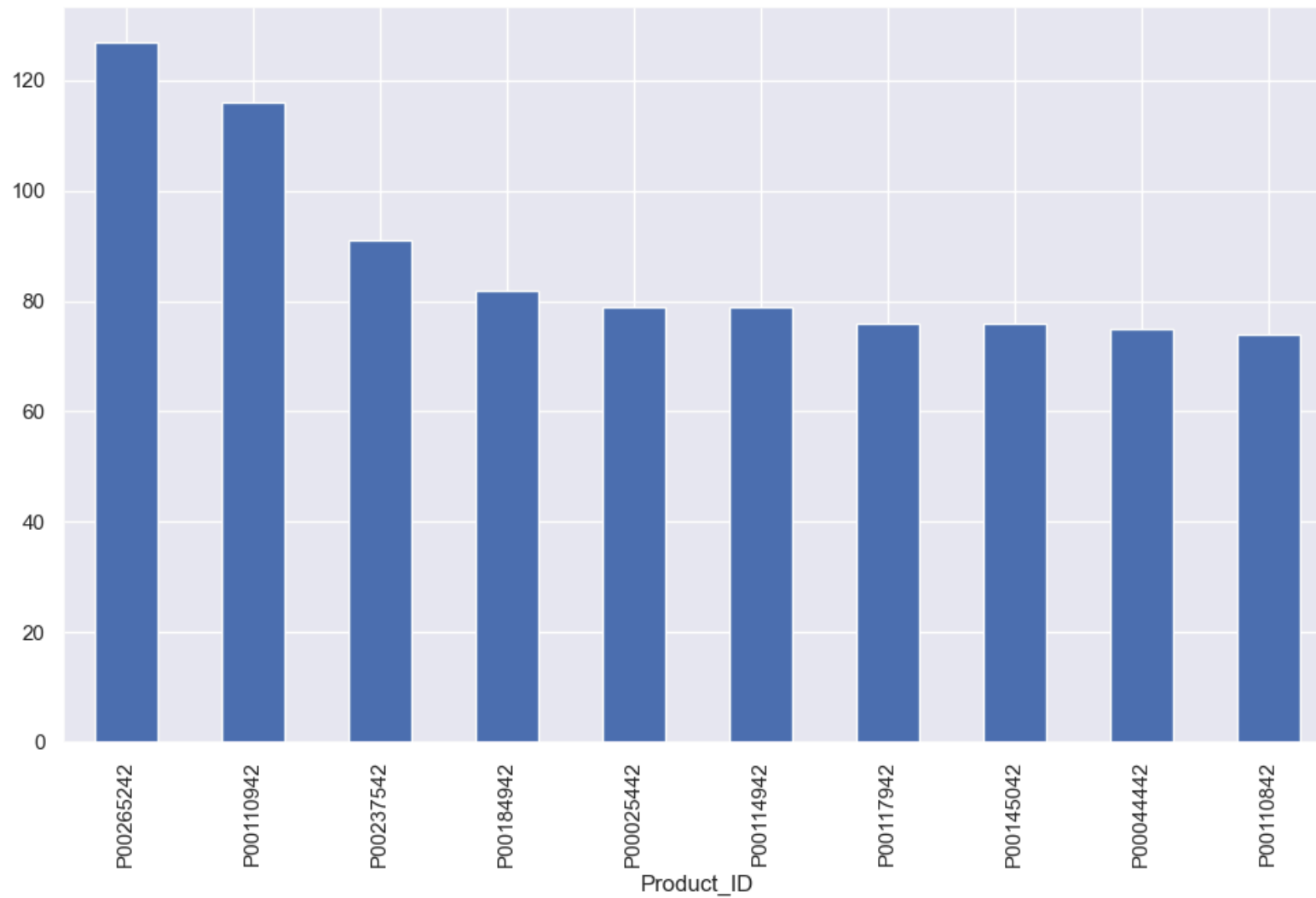
Out[27]: <Axes: xlabel='Product_ID', ylabel='Orders'>



In [28]: `# top 10 most sold products (same thing as above)`

```
fig1, ax1 = plt.subplots(figsize=(12,7))
df.groupby('Product_ID')['Orders'].sum().nlargest(10).sort_values(ascending=False).plot(kind='bar')
```

Out[28]: `<Axes: xlabel='Product_ID'>`



<h2 id="Conclusion:">Conclusion:</h2> <h3 id=""></h3>

<p>Married women age group 26-35 yrs from UP, Maharastra and Karnataka working in IT, Healthcare and Aviation are more likely to buy products from Food, Clothing and Electronics category</p>

<p>Feel free to connect and discuss with me: <https://www.linkedin.com/in/palak-tripathi-37a56234a/></p>

<p>Thank you!</p>